



Q1. What is CAT rating with respect to multimeters? The one I am buying indicates CAT III.

Dinesh Rai
Through E-mail

A1. All manufacturers of handheld multimeters are required to mention on their products the rated measurement category (CAT I, CAT II, CAT III or CAT IV). This marking is a convenient way for users to identify the maximum transient voltage that the device can safely withstand. Most handheld digital multimeters (DMMs) display this rating near the voltage/current input terminals. CAT III is the one you will generally find on the multimeters but beware some cheap Chinese manufacturers write CAT III just like that because they know what users are looking for.

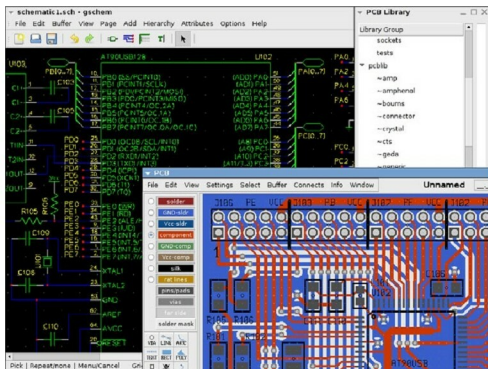
For example, 300V CAT II indicates that this module must not be connected to mains CAT II circuits when operated above 300V. Use the accompanying table to find out the impulse voltage level it can withstand. In this case, it will withstand impulse voltage up to 2500V.

CAT III category is applied to building circuit installations that are completely within the building, including parts of the service panel and the branch circuits. It also applies to many of the building's fixed equipment, which is connected directly to the building mains instead of being connected through cords and plugs.

Q2. Is hierarchical design really an important functionality for EDA tools?

Yogesh Shukla
Through E-mail

A2. This is a must have. When the design gets complex, it is better to represent it as a hierarchical design. This makes the representation



gEDA project tool

CAT Ratings and Corresponding Impulse Voltage Levels

	CAT IV	CAT III	CAT II	CAT I
150V	4000V	2500V	1500V	800V
300V	6000V	4000V	2500V	1500V
600V	8000V	6000V	4000V	2500V
1000V	12,000V	8000V	6000V	4000V
Resistance	2-ohm	2-ohm	12-ohm	30-ohm

simple, traceable and easy to navigate and review. In schematics with a lot of repetition, it can save your work. For example, if you have 16 channels of some circuitry, you can just build it once, and replicate it using hierarchy. Second, when multiple persons are working on a project, it enforces clean boundaries. Because all high-level connectivity is via ports, there is no risk of accidental connectivity via off-page connectors. We use gEDA project at EFY which supports hierarchical designs. Figure above shows a glimpse of gEDA project tool.

Q3. What is the difference between margin testing and verification?

Pooja Juyal
Through E-mail

A3. Engineers need to test their designs in order to ensure that they meet the design specifications across the full range of operation. This

is known as margin testing. This is a design challenge that requires the complete setup with stimulus and the acquisition systems. With such a system, you vary all the input parameters using the stimulus and check the limits till which you get an acceptable output. This testing also identifies how

much stress the device can take while still resuming its reliable operation.

Verification is required to check if a product meets a set of initial design requirements, specifications and regulations. In the development phase, verification procedures involve performing special tests on various modules and modelling the results, then performing a review or analysis of the modelled results to see if it complies with specifications. In the post-development phase, verification procedures involve regularly repeating tests devised specifically to ensure that the product continues to meet the initial design requirements, specifications and regulations as time progresses.

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